UNIVERSITÀ DEGLI STUDI DI TRIESTE

440MI DATA-DRIVEN SYSTEMS ENGINEERING

L01 - Introduction to MLOps

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ML Ops

L16 - Introduction to MLOps

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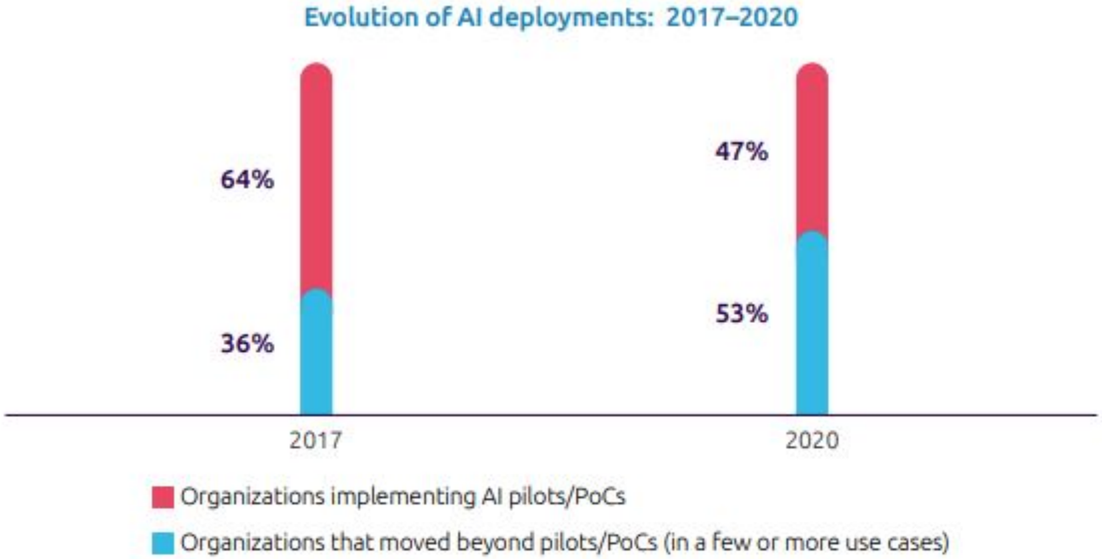
Today's goal:

- AI Maturity
- Overview of MLOps
- People and Roles
- Preparing for Production

Preliminaries:

Percentage of organizations that moved beyond AI pilots/pocs (Proof of Concept) increased to 53%



Source: Capgemini Research Institute, State of AI survey, March–April 2020, N=954 organizations implementing AI; State of AI survey, June 2017, N=993 organizations implementing AI.

Preliminaries:

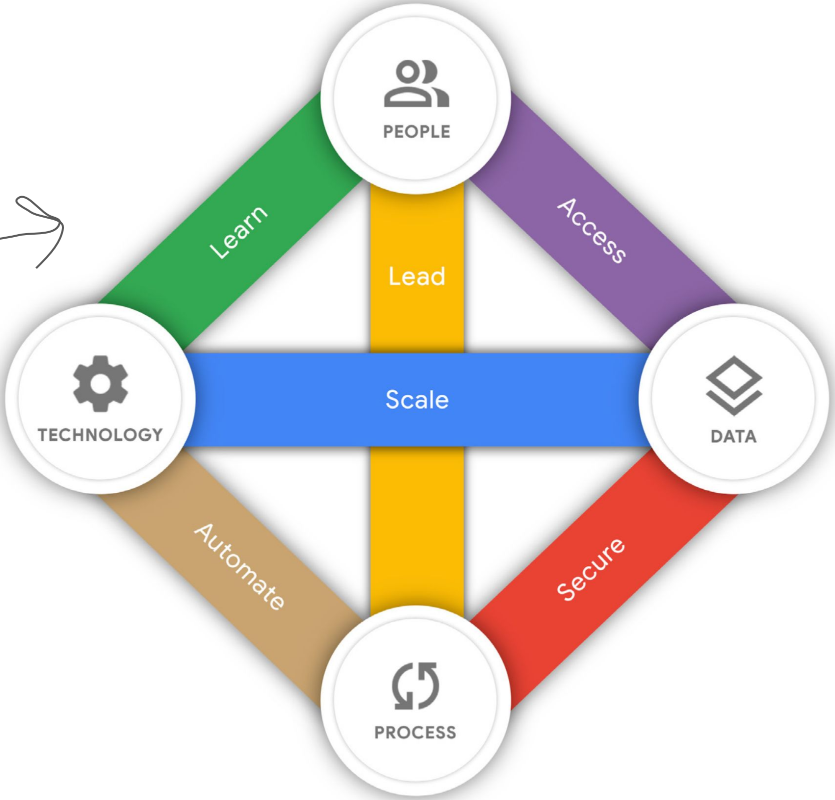
Why AI is so hard to be adopted?

1. Teams engage in a high degree of manual and one-off work.
2. They do not have reusable or reproducible components, and their processes involve difficulties in handoffs between data scientists and IT.
3. Lack of talent and integration issues as factors that can stall or derail AI initiatives.
4. Challenges in deployment, scaling, and versioning efforts still hinder teams from getting value from their investments in ML.
5. Challenges in achieving deployments at scale are lack of mid- to senior-level talent, lack of change-management processes, and lack of strong governance models for achieving scale.

Preliminaries: AI Maturity

- How do you take advantage of the power inherent in AI, while avoiding any potential missteps?

AI Maturity Themes!



Preliminaries: AI Maturity

For each theme, those practices will fall into one of the following phases:

The focus is on **easy** adoption, minimal disruption, and **quick** wins.

A broader vision governs AI adoption. Perception **starts to move** beyond the hype, **becoming** a pivotal accelerator for the **business**.

There is a **mechanism** in place for scaling and **promoting ML** capabilities across the organization.

Preliminaries: AI Maturity

Learn concerns the quality and scale of learning programs to upskill your staff, hire external talent, and augment your data science and ML engineering staff with experienced partners. What data and ML skill sets are required in the organization? What data science and engineering roles should be hired? To what extent do learning plans reflect business needs? What is the nature of the partnership with AI third parties?

Preliminaries: AI Maturity

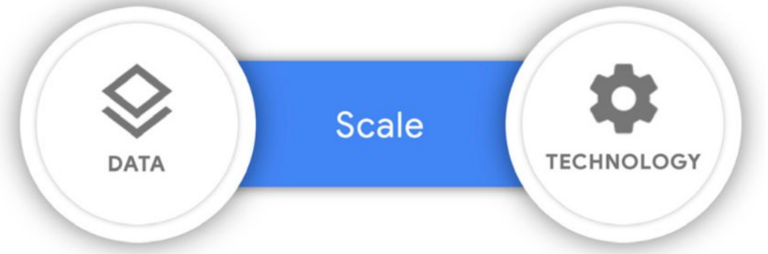
Lead concerns the extent to which your data scientists are supported by a mandate from leadership to apply ML to business use cases, and the degree to which the data scientists are cross-functional, collaborative, and self-motivated. How are the teams structured? Do they have executive sponsorship and empowerment? How are AI projects budgeted, governed, assessed?

Preliminaries: AI Maturity

Access concerns the extent to which your organization recognizes data management as a key element to enable AI and the degree to which data scientists can share, discover, and reuse data and other ML artifacts. How is the dataset created, curated, and annotated? Who owns the dataset? Is it discoverable and reusable? Can you share, reuse, and expand trained models, notebooks, and other ML components and solutions?

Preliminaries: AI Maturity

Scale concerns the extent to which you use cloud-native ML services that scale with large amounts of data and large numbers of data processing and ML jobs, with reduced operational overhead. How are cloud-based services provisioned? Are they on demand or long-living? How is capacity for workloads allocated?

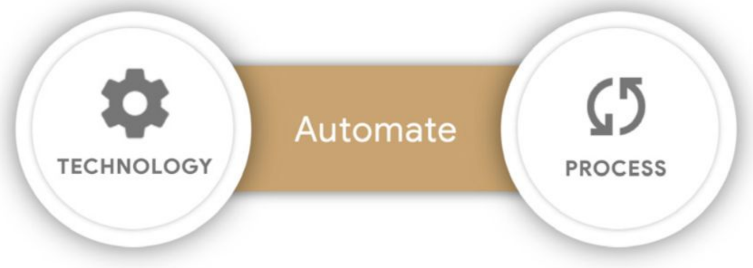


Preliminaries: AI Maturity

Secure concerns the extent to which you understand and protect your data and ML services from unauthorized and inappropriate access, in addition to ensuring responsible and explainable AI. What controls are in place? What strategies govern the whole? How does an organization establish trust in its AI capabilities so that it is leveraged to drive business value?

Preliminaries: AI Maturity

Automate concerns the extent to which you are able to deploy, execute, and operate technology for data processing and ML pipelines in production efficiently, frequently, and reliably. What triggers a process? How do you track data lineage? Are your pipelines fault tolerant and resumable? How do you manage logging, monitoring, and notifications?



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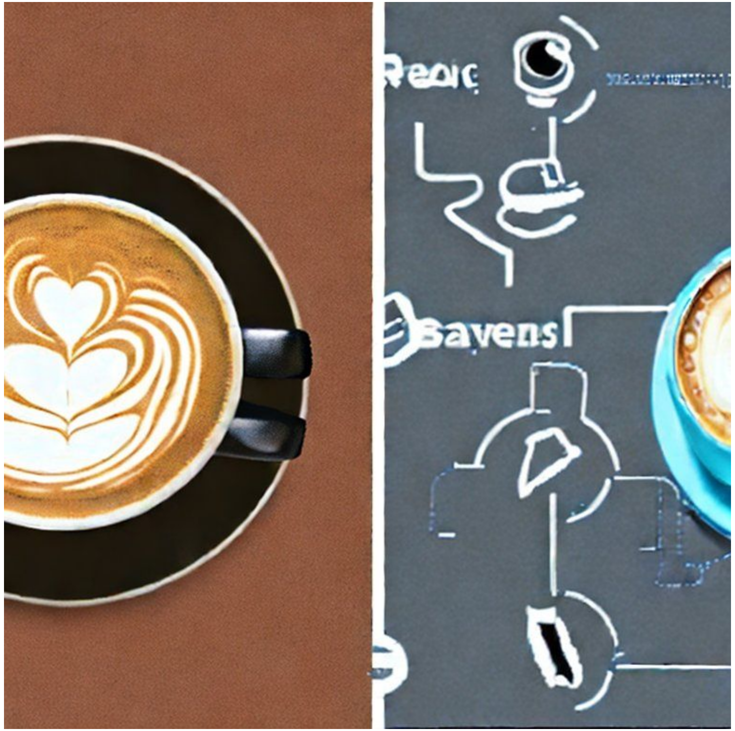
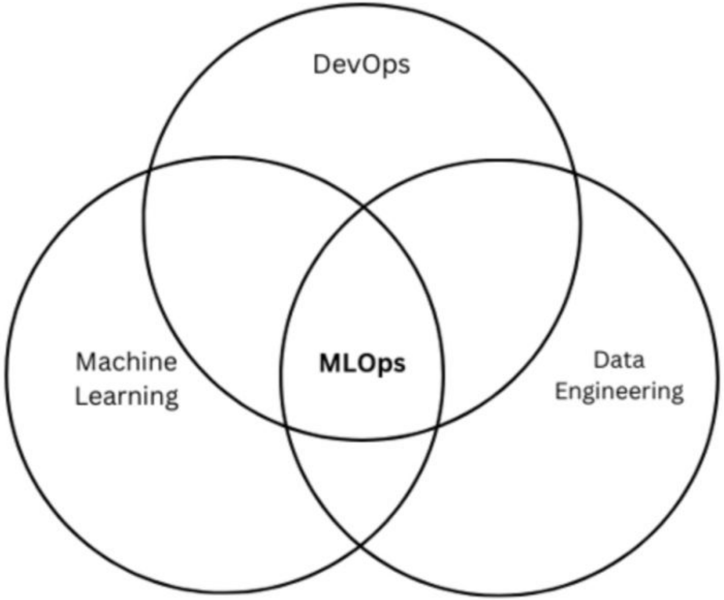


“AI maturity in a cup of coffee”

https://huggingface.co/spaces/stabilityai/stable-diffusion

Overview of MLOps



- MLOps is a **methodology** for ML engineering that unifies ML system development (the **ML element**) with ML system operations (the **Ops element**).
- It advocates **formalizing** and **automating** critical steps of ML system construction.
- MLOps **provides** a set of **standardized processes** and **technology** capabilities for building, deploying, and operationalizing ML systems rapidly and reliably.

Overview of MLOps

DevOps

When you deploy a web service, you care about:

- resilience;
- queries per second;
- load balancing;

MLOps

When you deploy an ML model, you also need to:

- worry about changes in the data;
- changes in the model;
- users trying to game the system;

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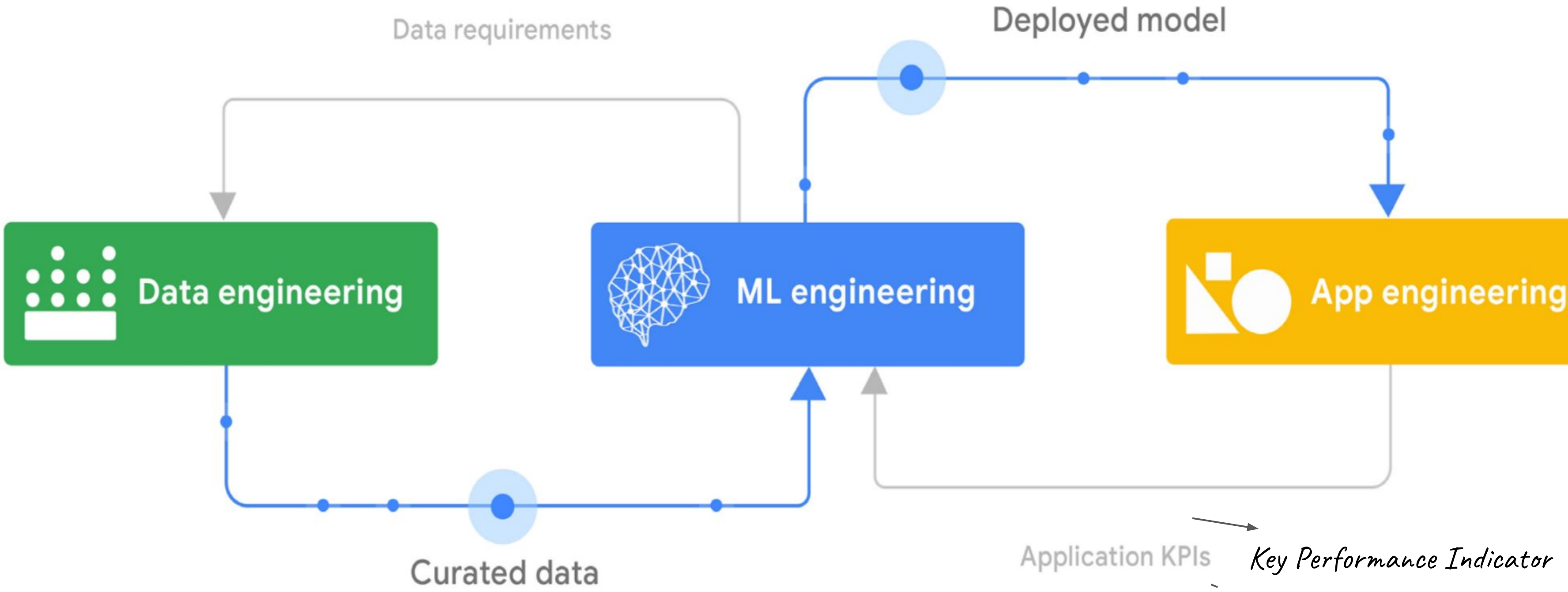
Overview of MLOps

Building ML-based System



Key Performance Indicator

Overview of MLOps

The MLOps lifecycle:

7 processes

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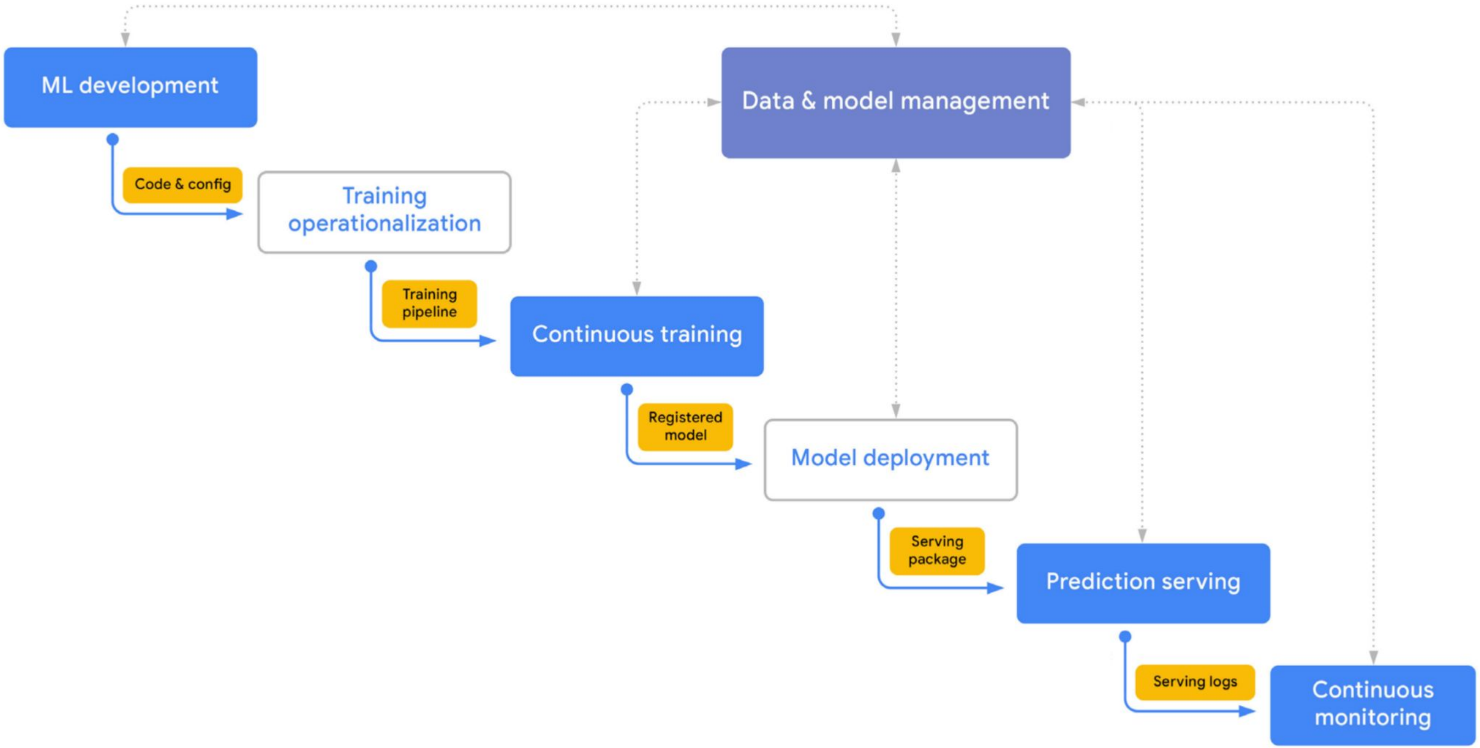
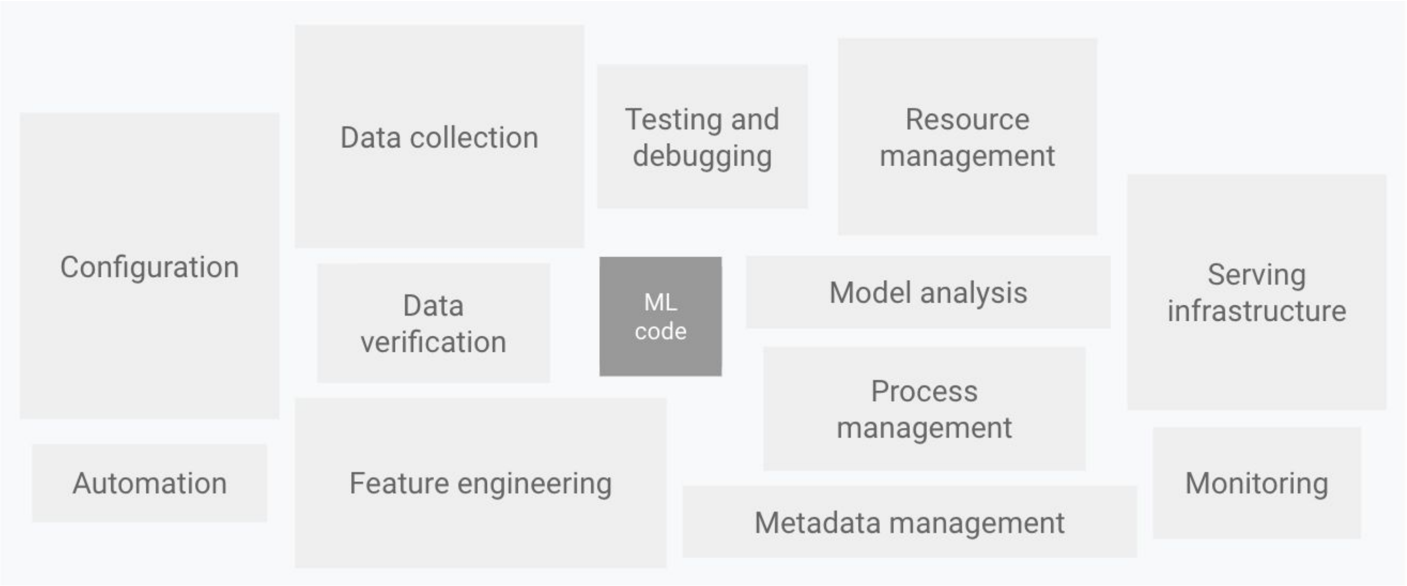
MLOps Workflow



MLOps Infrastructure



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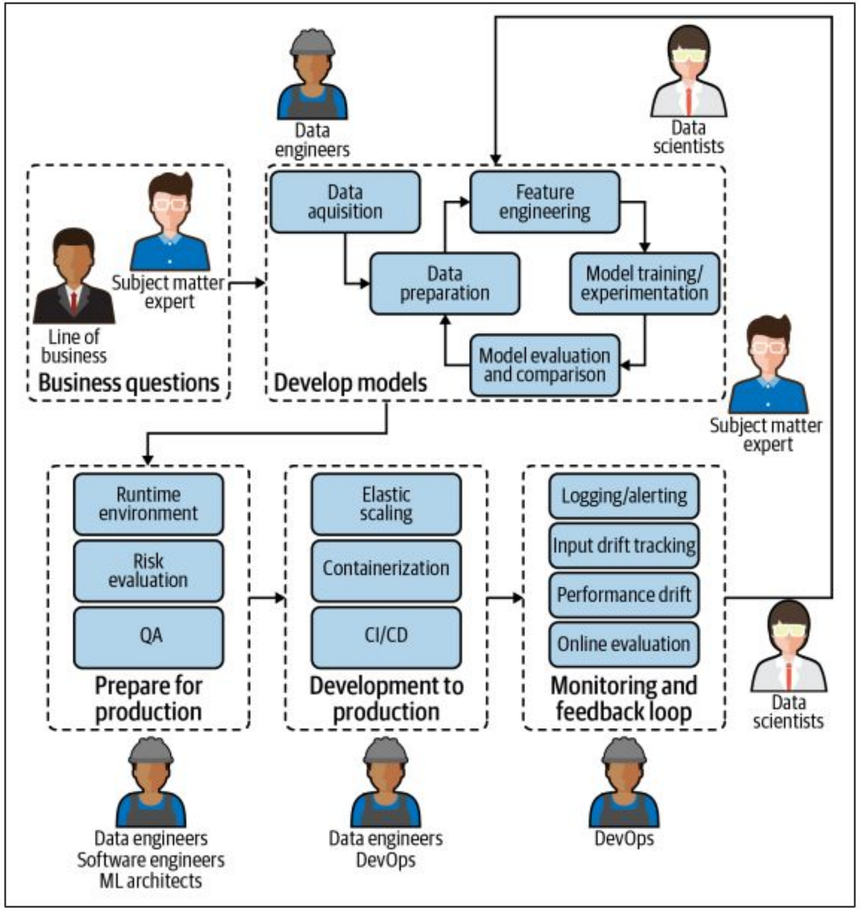
People and Roles:



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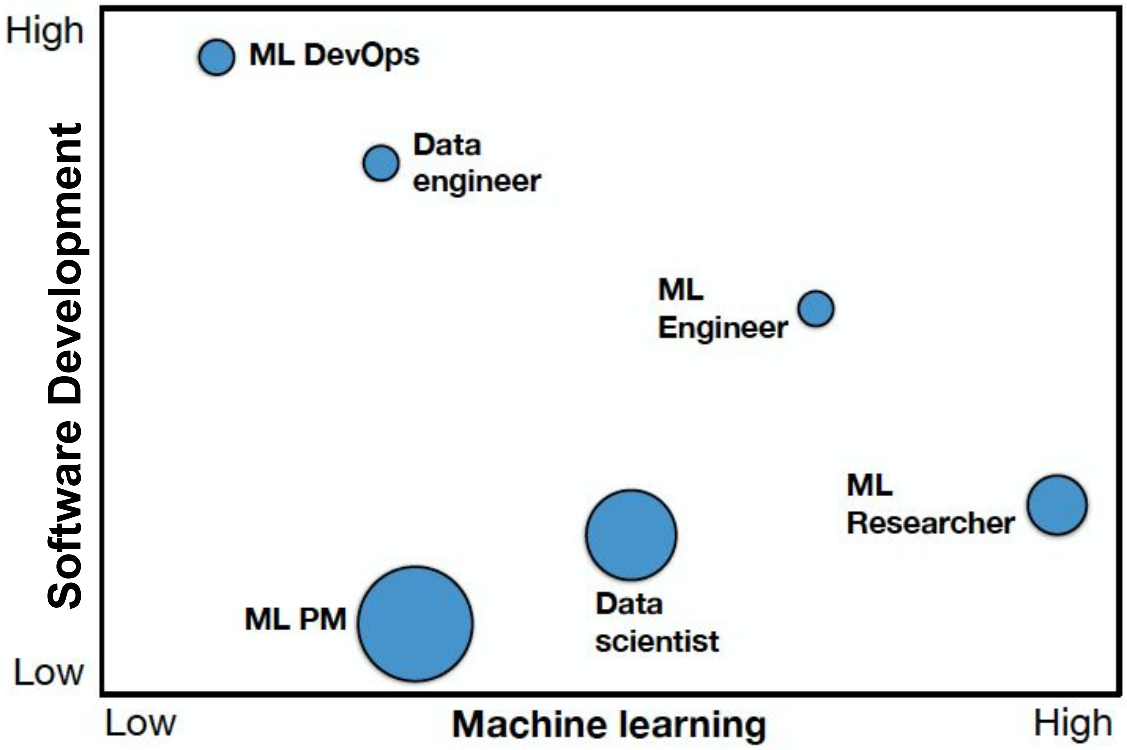
People and Roles:

AI (ML) and Dev Community!



People and Roles:

Subject matter experts:

Key Performance Indicator

Role:

- Provide business questions, **goals**, or **KPIs** around which ML models should be framed.
- **Continually evaluate** and ensure that model **performance** aligns with or resolves the initial need.

Requirements:

- Easy way to understand deployed model **performance in business terms**.
- Mechanism or feedback loop for model results.

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People and Roles:

Data scientists:

Role:

- **Build models** that address the business question or needs brought by subject matter experts.
- **Deliver operationalizable models** so that they can be properly used in the production environment and with production data.
- **Assess model quality** (of both original and tests) in tandem with subject matter experts to ensure they answer initial business questions or **needs.**

People and Roles:

Data scientists:

Requirements:

- **Automated model packaging** and **delivery** for quick and easy (yet safe) deployment to production.
- **Ability** to develop tests to **determine the quality of deployed models** and to make continual **improvements**.
- **Ability to investigate data pipelines** of each model to make quick assessments and adjustments regardless of who originally built the model.

People and Roles:

Data engineers:

Role:

- **Optimize** the **retrieval and use of data** to power ML models.

Requirements:

- **Visibility into performance** of all deployed models.
- **Ability to see** the full details of individual data pipelines to address underlying **data plumbing issues.**

People and Roles:

Software engineers:

Role:

- **Integrate ML models** in the company's **applications and systems**.
- **Ensure that ML models work** seamlessly with other non-machine-learning-based applications.

Requirements:

- **Versioning and automatic tests.**
- The ability to work in parallel on the same application.

People and Roles:

DevOps:

Role:

- Conduct and build operational systems and **test for security**, **performance**, **availability**.
- **Continuous Integration/Continuous Delivery** (CI/CD) pipeline management.

Requirements:

- Seamless integration of MLOps into the larger DevOps strategy of the enterprise.
- **Seamless deployment pipeline**.

People and Roles:

Machine learning architects:

Role:

- **Ensure a scalable and flexible** environment for **ML model pipelines**, from design to development and monitoring.
- **Introduce new technologies** when appropriate that **improve ML model** performance in production

Requirements:

- **High-level overview of models** and their resources consumed.
- **Ability to drill down into data pipelines to assess** and adjust infrastructure needs.

People and Roles:

Model risk managers/ auditors:

Role:

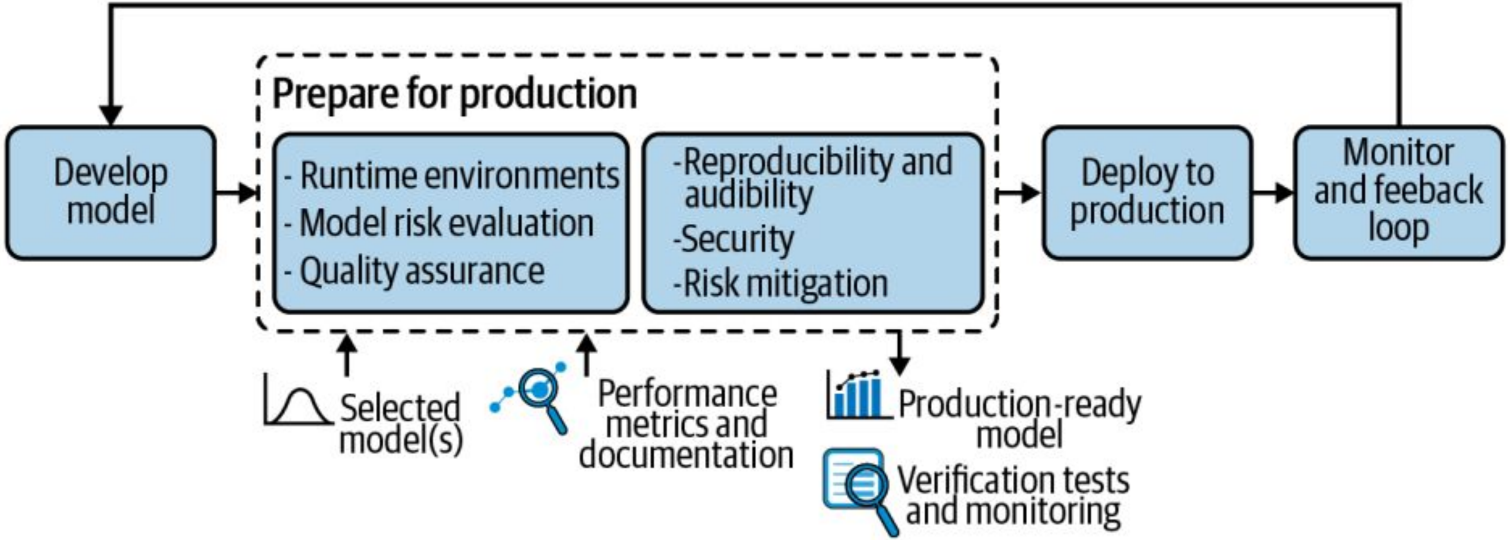
- **Minimize overall risk** to the company as a result of ML models in production.
- **Ensure compliance with internal and external requirements** before pushing ML models to production

Requirements:

- Robust, likely automated, **reporting tools** on all models (currently or ever in production), including data lineage.

Preparing for Production

- Confirming that something **works in the laboratory** has never been a sure sign it will work well in the real world, and machine learning models are no different.



How to Save and Load ML Models???

- **object serialization:**

  This process of saving a ML Model is also known as object serialization - representing an object with a stream of bytes, in order to store it on disk, send it over a network or save to a database.

- **deserialization:**

  While the restoring/reloading of ML Model procedure is known as deserialization.

- Common ways:

  1) Pickle Approach

  2) Joblib Approach

  3) JSON

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Textbook